

Subject: Computer Science Theory**Time allowed:** 2 hours**Maximum marks:** 50**Date:** Practice paperName: _____

Instructions

- All questions are compulsory.
- The marks for each question are shown in brackets on the right.
- Write all your answers in the answer sheet provided, not on this paper.
- Show your working for conversions, truth tables and traces — the working carries marks.

Part A — Multiple Choice Questions*8 questions — 8 marks*

Tick (✓) the correct option for each question.

Q1. What is the full form of URL?

[1]

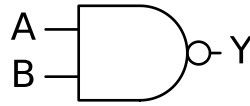
- (a) Uniform Resource Locator
 - (b) Universal Reference Link
 - (c) Uniform Retrieval Language
 - (d) User Resource Locator
-

Q2. Which of the following **is** a valid IP address?

[1]

- (a) 256.100.50.1
 - (b) 192.168.10.7
 - (c) 10.0.0
 - (d) 1.2.3.4.5
-

Q3. Which of the following is the Boolean expression for the output Y of the circuit below? [1]



- (a) $Y = \overline{A} \cdot \overline{B}$
 - (b) $Y = \overline{A} + \overline{B}$
 - (c) $Y = A \cdot B$
 - (d) $Y = A + B$
-

Q4. Which keyboard shortcut selects everything in the document? [1]

- (a) Ctrl + A
 - (b) Ctrl + E
 - (c) Ctrl + S
 - (d) Ctrl + Z
-

Q5. A picture stores 4 bits for each pixel. **How many different** colours can one pixel have? [1]

- (a) 4
 - (b) 8
 - (c) 16
 - (d) 32
-

Q6. You type =5+3*2 into a cell in LibreOffice Calc and press Enter. What appears in the cell? [1]

- (a) 16
 - (b) 11
 - (c) 13
 - (d) =5+3*2
-

Q7. You want to email a report so that it looks exactly the same on every computer that opens it. Which format should you use? [1]

- (a) .txt
 - (b) .pdf
 - (c) .zip
 - (d) .bmp
-

Q8. What is the value of $A + \overline{A}$? [1]

- (a) A
 - (b) $\overline{A}$
 - (c) 0
 - (d) 1
-

Part B — Number Systems

2 questions — 5 marks

Q9. Convert the decimal number 89_{10} to binary. [2]

Q10. Convert the binary number 110110111_2 to hexadecimal. [3]

Part C — Boolean Algebra

2 questions — 6 marks

Q11. For the Boolean expression below: [3]

$$Y = \overline{A} \cdot B \cdot \overline{C} + A \cdot C$$

- (i) Draw the complete truth table for Y .
 - (ii) Find the value of Y when $A = 1, B = 1, C = 1$ and when $A = 0, B = 1, C = 0$. Show the substitution.
-

Q12. For the Boolean expression below: [3]

$$Y = \overline{A \cdot B} + C$$

- (i) Draw the logic-gate circuit for Y .
 - (ii) Draw the truth table for the same circuit.
-

Q13. Draw the logic-gate circuit for the Boolean expression below. [1]

$$Y = \overline{A} + B \cdot C$$

Q14. The shaded cells in the spreadsheet below are selected. Write the address of this selection. [1]

	A	B	C	D	E	F
1						
2						
3						
4						
5						
6						

Q15. What is the keyboard shortcut for saving a document in LibreOffice Writer? [1]

Q16. A black-and-white picture uses one **bit** for each pixel. How many bits are needed for a picture that is **8 pixels by 8 pixels**? [1]

Q17. What is the value of `31 // 7 - 31 % 7`? [1]

Q18. Which of the following is an example of system software? [1]

- (a) A photo editor
- (b) A web browser
- (c) An operating system
- (d) A spreadsheet program

Q19. The spreadsheet below is open in LibreOffice Calc. Write what appears in each cell after the formula is typed into it and Enter is pressed. [2]

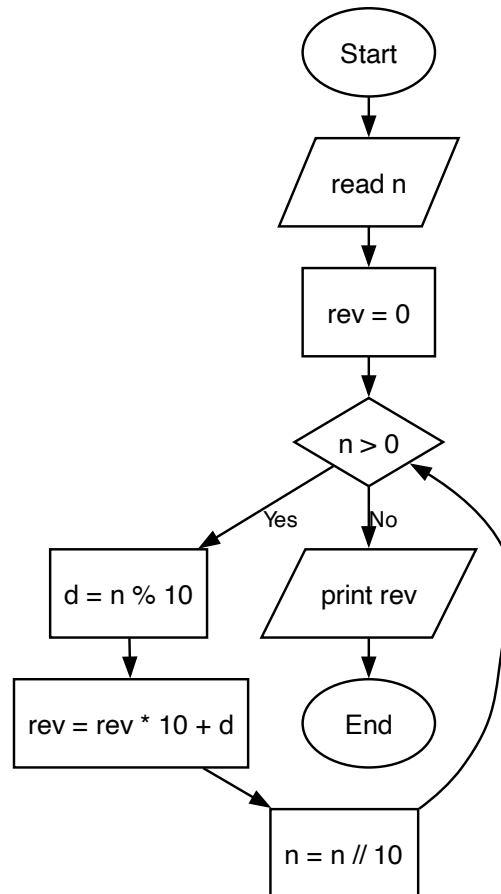
	A	B	C	D
1	Student	Marks		
2	Asha	72		
3	Bilal	45		
4	Chen	60		
5	Dia	88		
6	Ekta	35		
7	Farid	96		
8				
9				

(i) Cell B8 holds the formula =SUM(B2:B7). What does B8 show?

(ii) Cell B9 holds the formula =SUMIF(B2:B7,">=60"). What does B9 show?

Q20. Trace this algorithm for $n = 5309$, and draw the trace table.

[5]

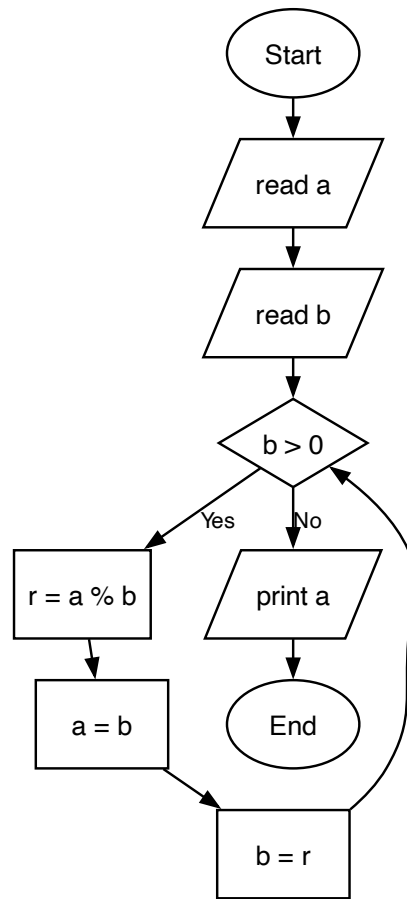


Copy this table into your answer sheet and complete it.

n	rev	d	output

Q21. Trace this algorithm for $a = 120, b = 45$, and draw the trace table.

[4]

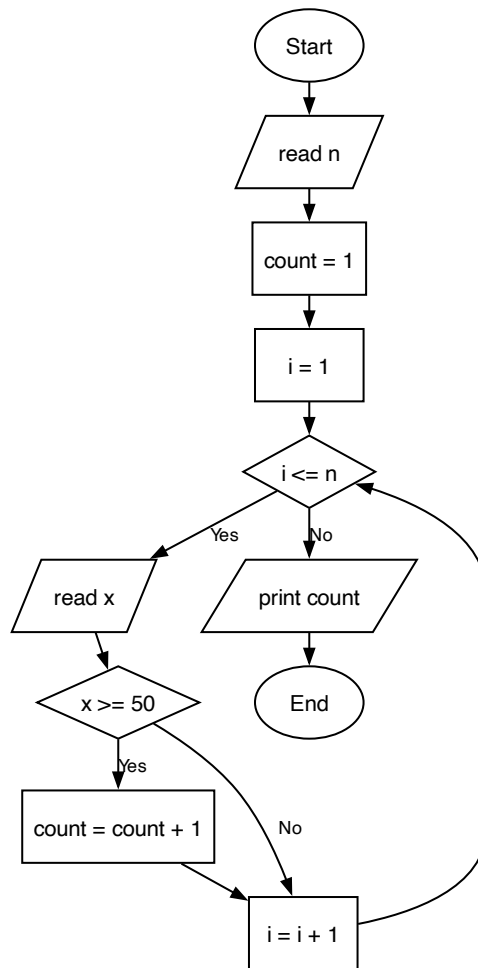


Copy this table into your answer sheet and complete it.

a	b	r	output

Q22. The flowchart below is **meant** to do what the italic line says, but there are mistakes in it. The table under the flowchart shows what it should print and what it actually prints. **Find all the mistakes in this flowchart, then draw the corrected flowchart in your answer sheet**, so that it gives the correct expected output. [2]

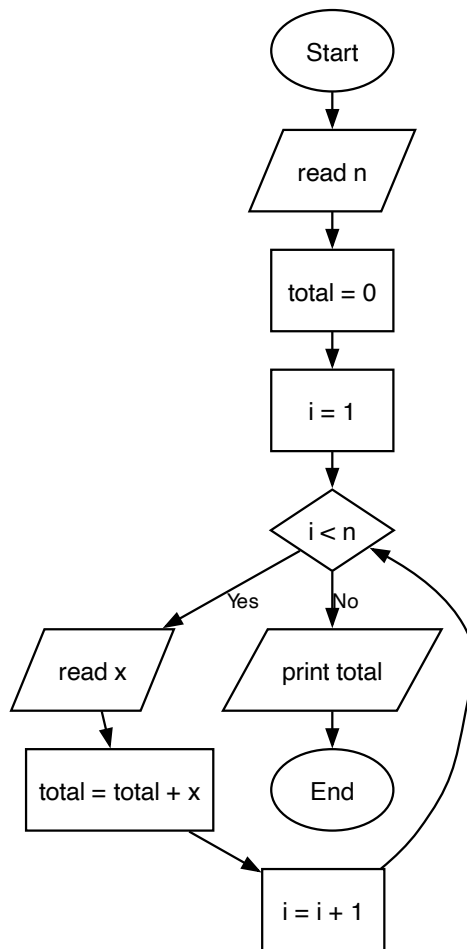
The program should read how many marks there are, then read that many marks, and finally print how many of them are more than 50.



input	should print	actually prints
n = 5, x = 72, 50, 45, 90, 60	3	5

Q23. The flowchart below is **meant** to do what the italic line says, but there are mistakes in it. The table under the flowchart shows what it should print and what it actually prints. **Find all the mistakes in this flowchart, then draw the corrected flowchart in your answer sheet**, so that it gives the correct expected output. [2]

The program should read how many numbers there are, then read that many numbers, and print their total.



input	should print	actually prints
n = 4, x = 10, 20, 30, 40	100	60
n = 5, x = 5, 15, 25, 35, 45	125	80

Q24. Read the description below, then answer the parts that follow.

[7]

A program reads a whole number n and prints every factor of n — that is, every whole number from 1 up to n that divides into n leaving no remainder. For example, the factors of 6 are 1, 2, 3 and 6.

- (i) Write the algorithm in words, step by step.
 - (ii) Draw the flowchart for that algorithm.
 - (iii) Trace your flowchart for $n = 12$ and $n = 7$, and write what it prints in each case.
-

Q25. Read the description below, then answer the parts that follow.

[3]

An electricity board bills a house. The program reads the number of units of electricity used, and works out the bill as $\text{units} \times 8$ rupees. If more than 100 units were used, a fixed surcharge of Rs 500 is added to the bill. The program then prints the amount the house has to pay.

- (i) Write the algorithm in words, step by step.
- (ii) Draw the flowchart for that algorithm.